

t -STRUCTURES

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The purpose of these notes is to concisely define and discuss basic properties of t -structures and the perverse t -structure of complexes of constructible ℓ -adic sheaves

1. TRIANGULATED CATEGORIES

Definition 1.0.1 (Triangulated category). A *triangulated category* is a triple $(\mathcal{T}, [1], \Delta)$ where:

- $\mathcal{T}$ is an additive category,
- $[1] : \mathcal{T} \rightarrow \mathcal{T}$ (also denoted by notations such as Σ or T) is an additive auto-equivalence called the *shift* or *suspension functor*,
- Δ is a class of *distinguished triangles* of the form

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

with objects $X, Y, Z \in \mathcal{T}$ and morphisms f, g, h in $\mathcal{T}$,

satisfying the following axioms:

(TR1) The class Δ is closed under isomorphisms of triangles and contains the triangles isomorphic to

$$A \xrightarrow{\text{id}} A \rightarrow 0 \rightarrow A[1].$$

Moreover, for every morphism $f : X \rightarrow Y$ in $\mathcal{T}$, there exists a distinguished triangle

$$X \xrightarrow{f} Y \rightarrow Z \rightarrow X[1].$$

(TR2) (Rotation) If

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

is a distinguished triangle, then so are

$$Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-f[1]} Y[1]$$

and

$$Z[-1] \xrightarrow{-h[-1]} X \xrightarrow{f} Y \xrightarrow{g} Z.$$

(TR3) (Octahedral axiom) Given morphisms $X \xrightarrow{f} Y$ and $Y \xrightarrow{g} Z$ in $\mathcal{T}$, there exist distinguished triangles

$$X \xrightarrow{f} Y \xrightarrow{u} C(f) \xrightarrow{v} X[1],$$

$$Y \xrightarrow{g} Z \xrightarrow{w} C(g) \xrightarrow{x} Y[1],$$

$$X \xrightarrow{g \circ f} Z \xrightarrow{y} C(g \circ f) \xrightarrow{z} X[1],$$

and morphisms

$$s : C(f) \rightarrow C(g \circ f), \quad t : C(g \circ f) \rightarrow C(g)$$

such that the following diagram commutes and its rows and columns are distinguished triangles:

$$\begin{array}{ccccccc}
X & \xrightarrow{f} & Y & \xrightarrow{u} & C(f) & \xrightarrow{v} & X[1] \\
\parallel & & \downarrow g & & \downarrow s & & \parallel \\
X & \xrightarrow{g \circ f} & Z & \xrightarrow{y} & C(g \circ f) & \xrightarrow{z} & X[1] \\
& & \downarrow w & & \downarrow t & & \\
& & C(g) & \xlongequal{\quad} & C(g) & & \\
& & \downarrow x & & \downarrow & & \\
& & Y[1] & \xrightarrow{f[1]} & X[1][1] & &
\end{array}$$

In particular,

- The compositions $s \circ u = y$, $t \circ s = w$, and $x \circ t = v[1]$ hold,
- All rows and columns form distinguished triangles.

(TR4) The class Δ is closed under the shift functor $[1]$: if

$$X \rightarrow Y \rightarrow Z \rightarrow X[1]$$

is distinguished, then so is

$$X[1] \rightarrow Y[1] \rightarrow Z[1] \rightarrow X[2].$$

Definition 1.0.2. Let $\mathcal{T}$ be a triangulated category (Definition 1.0.1). For any morphism $u : X \rightarrow Y$ in $\mathcal{T}$, a **cone of u** is an object C that fits into a distinguished triangle (Definition 1.0.1):

$$X \xrightarrow{u} Y \xrightarrow{v} C \xrightarrow{d} X[1]$$

In a general triangulated category, the cone C is guaranteed to exist by the axioms, but it is only unique up to a (not necessarily unique) isomorphism.

Definition 1.0.3. Let $\mathcal{T}$ be a triangulated category (Definition 1.0.1). Let (X, Y, Z, u, v, d) and (X', Y', Z', u', v', d') be two distinguished triangles (Definition 1.0.1) in $\mathcal{T}$. A **morphism of triangles** is a triple (f, g, h) of morphisms $f : X \rightarrow X'$, $g : Y \rightarrow Y'$, and $h : Z \rightarrow Z'$ such that the following diagram commutes:

$$\begin{array}{ccccccc} X & \xrightarrow{u} & Y & \xrightarrow{v} & Z & \xrightarrow{d} & X[1] \\ \downarrow f & & \downarrow g & & \downarrow h & & \downarrow f[1] \\ X' & \xrightarrow{u'} & Y' & \xrightarrow{v'} & Z' & \xrightarrow{d'} & X'[1] \end{array}$$

That is, the relations $g \circ u = u' \circ f$, $h \circ v = v' \circ g$, and $f[1] \circ d = d' \circ h$ must all hold.

Definition 1.0.4. A functor $T : D_1 \rightarrow D_2$ between triangulated categories (Definition 1.0.1) is called **exact** if it is additive, is translation preserving, and transforms distinguished triangles (Definition 1.0.1) to distinguished triangles. Oftentimes, a **morphism between triangulated categories** or a **triangulated functor** refers to an exact functor between triangulated categories.

Proposition 1.0.5 ([BBD82, Proposition 1.1.9.]). Let $\mathcal{T}$ be a triangulated category (Definition 1.0.1). Let (X, Y, Z, u, v, d) and (X', Y', Z', u', v', d') be two distinguished triangles (Definition 1.0.1) in $\mathcal{T}$. Let $g : Y \rightarrow Y'$ be a morphism in $\mathcal{T}$. Consider the following commutative diagram:

$$\begin{array}{ccccccc} X & \xrightarrow{u} & Y & \xrightarrow{v} & Z & \xrightarrow{d} & X[1] \\ \downarrow f & & \downarrow g & & \downarrow h & & \downarrow f[1] \\ X' & \xrightarrow{u'} & Y' & \xrightarrow{v'} & Z' & \xrightarrow{d'} & X'[1] \end{array}$$

The following conditions are equivalent:

- (a) The composition $v' \circ g \circ u = 0$.
- (b) There exists a morphism $f : X \rightarrow X'$ such that $g \circ u = u' \circ f$.
- (b') There exists a morphism $h : Z \rightarrow Z'$ such that $h \circ v = v' \circ g$. (♠ TODO: morphism of distinguished triangles)
- (c) There exists a morphism of triangles (f, g, h) from (X, Y, Z) to (X', Y', Z') .

If these conditions are verified, and if $\text{Hom}_{\mathcal{T}}(X, Z'[-1]) = 0$, the morphism f (resp. h) of (b) (resp. (b')) is unique.

Corollary 1.0.6 ([BBD82, Corollary 1.1.10.]). Let $\mathcal{T}$ be a triangulated category (Definition 1.0.1). Let $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{d} X[1]$ be a distinguished triangle (Definition 1.0.1) in $\mathcal{T}$. If $\text{Hom}_{\mathcal{T}}(X, Z[-1]) = 0$, then:

- (i) the cone of u (Definition 1.0.2) is unique up to unique isomorphism;
- (ii) d is the unique morphism $x : Z \rightarrow X[1]$ such that the triangle $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{x} X[1]$ is distinguished.

2. t -STRUCTURES

See [BBD82, 1.3], [KW13, II.2-4, III.1].

D : A triangulated category.

T : The translation functor $T : D \rightarrow D$.

$A[1]$: $T(A)$ for $A \in D$.

Definition 2.0.1. A *strictly full subcategory of a category* is a subcategory that is full (Definition A.0.2) and also closed under isomorphism.

Definition 2.0.2 ([KW13, Definition II.2.1]). Let D be a triangulated category (Definition 1.0.1).

A t -structure on D consists of two strictly full subcategories (Definition 2.0.1) $D^{\leq 0}$ and $D^{\geq 0}$ of D such that, writing

$$\begin{aligned} D^{\leq n} &= D^{\leq 0}[-n] \\ D^{\geq n} &= D^{\geq 0}[-n] \end{aligned}$$

we have

1. $\text{Hom}(D^{\leq 0}, D^{\geq 1}) = 0$,
2. $D^{\leq 0} \subset D^{\leq 1}$ and $D^{\geq 1} \subset D^{\geq 0}$, and
3. for every object E in D there exists a distinguished triangle (Definition 1.0.1) (A, E, B) with $A \in D^{\leq 0}$ and $B \in D^{\geq 1}$.

A t -category is a triangulated category equipped with a t -structure.

Given a t -structure t on D , the *core of t* is the full subcategory $\text{Core}(D) = \text{Core}_t(D) := D^{\leq 0} \cap D^{\geq 0}$ of D . It is an abelian category (Theorem 2.0.13).

(♠ **TODO:**) From now on, let $D^{\leq 0}, D^{\geq 0}$ be a t -structure on D unless otherwise specified.

¹A *strictly full subcategory of a category* is a subcategory that is full and also closed under isomorphism.

Lemma 2.0.3. Let D be a triangulated category (Definition 2.0.2) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D . If $m, n \in \mathbb{Z}$ satisfy $m \leq n$, then

$$D^{\leq m} \supset D^{\leq n}$$

$$D^{\geq m} \subset D^{\geq n}$$

Proof. This is clear from the definitions and axioms on $D^{\leq n}, D^{\geq n}$. $\square$

Definition 2.0.4. Given a triangulated category (Definition 1.0.1) D with a t -structure (Definition 2.0.2) $(D^{\leq 0}, D^{\geq 0})$, call an object $M \in D$ *bounded* if $M \in D^{\geq a} \cap D^{\leq b}$ for some integers a and b . Say that D is *bounded with respect to the t -structure* if every object of D is bounded.

Proposition 2.0.5 ([BBD82, 1.3.3(i)]). Let D be a triangulated category (Definition 2.0.2) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D .

For all $n \in \mathbb{Z}$, the inclusion functor $D^{\leq n} \hookrightarrow D$ has a right adjoint (Definition A.0.1) $\tau_{\leq n} : D \rightarrow D^{\leq n}$ and the inclusion functor $D^{\geq n} \hookrightarrow D$ has a left adjoint (Definition A.0.1) $\tau_{\geq n} : D \rightarrow D^{\geq n}$.

Lemma 2.0.6. Let D be a triangulated category (Definition 2.0.2) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D .

We have

$$\begin{aligned}\tau_{\leq n+1}(X) &= (\tau_{\leq n}(X[1]))[-1] \\ \tau_{\geq n+1}(X) &= (\tau_{\geq n}(X[1]))[-1].\end{aligned}$$

Equivalently,

$$\begin{aligned}\tau_{\leq n}(X) &= (\tau_{\leq 0}(X[n]))[-n] \\ \tau_{\geq n}(X) &= (\tau_{\geq 0}(X[n]))[-n].\end{aligned}$$

for all integer n .

Proof. Note that for $n = 0$, these are trivially true. Inductively suppose that these are true for $n = m$. We then have

$$\begin{aligned}\tau_{\leq m+1}(X) &= \tau_{\leq m}(X[1])[-1] = (\tau_{\leq 0}(X[n+1])[-n])[-1] = \tau_{\leq 0}(X[n+1])[-n-1] \\ \tau_{\geq m+1}(X) &= \tau_{\geq m}(X[1])[-1] = (\tau_{\geq 0}(X[n+1])[-n])[-1] = \tau_{\geq 0}(X[n+1])[-n-1]\end{aligned}$$

and hence the desired equalities holds for all $n \geq 0$ by induction. Similar steps demonstrate the desired equalities for $n \leq 0$ as well. $\square$

From here on out, given a t -structure, fix such functors $\tau_{\leq 0}$ and $\tau_{\geq 1}$.

Proposition 2.0.7 ([BBD82, Proposition 1.3.3(ii)]). Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D .

For any object E in D , there exists a unique morphism $d \in \text{Hom}(\tau_{\geq 1}E, \tau_{\leq 0}E[1])$ such that the triangle

$$\tau_{\leq E} \rightarrow E \rightarrow \tau_{\geq 1}E \xrightarrow{d} \tau_{\leq 0}E[1]$$

is distinguished (Definition 1.0.1). Up to unique isomorphism, every distinguished triangle (A, E, B) in D with $A \in D^{\leq 0}$ and $B \in D^{\geq 1}$ is equivalent to such a triangle.

Lemma 2.0.8. Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D .

We have

$$\begin{aligned} \tau_{\leq m} \circ \tau_{\leq n} &= \tau_{\leq m} & \text{for } m \leq n \\ \tau_{\geq m} \circ \tau_{\geq n} &= \tau_{\geq n} & \text{for } m \leq n. \end{aligned}$$

Proof. We prove the first statement. By Yoneda's lemma, it suffices to show that there are natural isomorphisms

$$\text{Hom}_{D^{\leq m}}(A, \tau_{\leq m}\tau_{\leq n}B) \cong \text{Hom}_{D^{\leq m}}(A, \tau_{\leq m}B)$$

for all objects $A \in D^{\leq m}$ and $B \in D$. By Proposition 2.0.5, the left hand side is isomorphic to $\text{Hom}_D(A, \tau_{\leq n}B)$. As $A \in D^{\leq m} \subset D^{\leq n}$, $B \in D^{\leq n}$, and $D^{\leq n}$ is a full subcategory of D , the Hom-group $\text{Hom}_D(A, \tau_{\leq n}B)$ is identifiable with the Hom-group $\text{Hom}_{D^{\leq n}}(A, \tau_{\leq n}B)$. In turn, this Hom-group is identifiable with $\text{Hom}_D(A, B)$, which is identifiable with $\text{Hom}_{D^{\leq m}}(A, \tau_{\leq m}B)$ as desired by the adjunction of $\tau_{\leq n}$.

□

Lemma 2.0.9 (See [KW13, Corollary II.1.6]). Let D be a triangulated category (Definition 1.0.1). Let (X, Y, Z, u, v, w) be a distinguished triangle (Definition 1.0.1) of D . The morphism u is an isomorphism if and only if $Z \cong 0$.

Lemma 2.0.10 (Orthogonality, see [KW13, Lemma II.2.2]). Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D . Let $E \in D$. The following are equivalent:

1. $E \in D^{\geq n+1}$
2. $\text{Hom}(D^{\leq n}, E) = 0$.

The following are equivalent:

1. $E \in D^{\leq n-1}$
2. $\text{Hom}(E, D^{\geq n}) = 0$.

Proof. We show the former equivalence for $n = 0$ — the dual equivalence and both equivalences for general n can be easily deduced. Note that (2) immediately implies (1) by the

axioms of a t -structure. Conversely, we first deduce that $\tau_{\leq 0}(E) = 0$ by the adjunction isomorphism and Lemma 2.0.8:

$$\mathrm{Hom}_{D^{\leq 0}}(\tau_{\leq 0}(E), \tau_{\leq 0}(E)) \cong \mathrm{Hom}_D(\tau_{\leq 0}\tau_{\leq 0}(E), E) \subset \mathrm{Hom}_D(\tau_{\leq 0}(E), E) = 0.$$

Taking the distinguished triangle

$$\tau_{\leq 0}(E) \rightarrow E \rightarrow \tau_{\geq 1}(E) \rightarrow \tau_{\leq 0}(E)[1]$$

of Proposition 2.0.7 and rotating it appropriately, Claim 2.0.9 shows that $E \cong \tau_{\geq 1}(E)$ and hence $E \in D^{\geq n+1}$. $\square$

Lemma 2.0.11 (Extensions, see [KW13, Lemma II.2.3]). Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D .

Let (X, Y, Z) be a distinguished triangle. then

1. If $X, Z \in D^{\leq n}$, then $Y \in D^{\leq n}$ as well.
2. If $Y \in D^{\leq n}$ and if $X \in D^{\leq n+1}$, then $Z \in D^{\leq n}$.

Proof. (♠ TODO: TODO: prove) The long exact sequence

$$\begin{aligned} \cdots \rightarrow \mathrm{Hom}(X[n+1], A) \rightarrow \mathrm{Hom}(Z[n], A) \rightarrow \mathrm{Hom}(Y[n], A) \rightarrow \mathrm{Hom}(X[n], A) \rightarrow \cdots \\ \cdots \rightarrow \mathrm{Hom}(A, X[n]) \rightarrow \mathrm{Hom}(A, Y[n]) \rightarrow \mathrm{Hom}(A, Z[n]) \rightarrow \mathrm{Hom}(A, X[n+1]) \rightarrow \cdots \end{aligned}$$

$\square$

Lemma 2.0.12 (Compatibility, see [KW13, Lemma II.2.4]). Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D .

We have

$$\begin{aligned} \tau_{\geq m}(D^{\leq n}) &\subset D^{\leq n} \\ \tau_{\leq m}(D^{\geq n}) &\subset D^{\geq n} \end{aligned}$$

(Proposition 2.0.5) for all $m, n \in \mathbb{Z}$. We have

$$\tau_{\geq m}(D^{\leq n}) = \tau_{\leq n}(D^{\geq m}) = 0$$

for $m > n$.

Proof. We prove the first inclusion. For $n < m$ and $X \in D^{\leq n}$ the unit map $X \rightarrow \tau_{\geq m}X$ of the adjunction for $\tau_{\geq m}$ is zero as a consequence of the axioms of a t -structure. The adjunction then implies that $\mathrm{Hom}(\tau_{\geq m}X, \tau_{\geq m}X) = 0$, so $\tau_{\geq m}X = 0$.

Now suppose that $m \leq n$. If $X \in D^{\leq n}$, then $\tau_{\leq m-1}(X) \in D^{\leq m-1} \subset D^{\leq n+1}$. In the distinguished triangle $(\tau_{\leq m-1}X, X, \tau_{\geq m}X)$ and applying the extensions Lemma 2.0.11, we have that $\tau_{\geq m}X \in D^{\leq n}$.

Similarly, the second inclusion holds. Note that we already proved the vanishing. $\square$

Theorem 2.0.13 (See [KW13, Theorem II.3.1]). Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D . The core $\text{Core}(D)$ (Definition 2.0.2) is an abelian category. A sequence

$$0 \rightarrow X \xrightarrow{u} Y \xrightarrow{v} Z \rightarrow 0$$

is exact if and only if there exists a distinguished triangle (Definition 1.0.1) (X, Y, Z, u, v, w) in D .

3. COHOMOLOGY FUNCTORS FOR A t -STRUCTURE

Definition 3.0.1. Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D . Define the *cohomology functors* $H^n : D \rightarrow \text{Core}(D)$ as follows:

$$\begin{aligned} H^0(X) &= \tau_{\leq 0} \tau_{\geq 0} X \\ H^n(X) &= H^0(X[n]). \end{aligned}$$

Note that $H^0(X) \in \text{Core}(D)$ by Lemma 2.0.12.

Lemma 3.0.2. Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D . Let $X \in D$.

We have

1. $H^a(X[b]) = H^{a+b}(X) = H^0(X[a+b])$ (Definition 3.0.1) for all $a, b \in \mathbb{Z}$, and
2. $H^n(X) = (\tau_{\leq n} \tau_{\geq n} X)[n]$ (Proposition 2.0.5) (Definition 1.0.1) for all $n \in \mathbb{Z}$.

Proof. 1. Observe that

$$H^a(X[b]) = H^0((X[b])[a]) = H^0(X[a+b]) = H^{a+b}(X).$$

2. The left hand side equals

$$H^0(X[n]) = \tau_{\leq 0} \tau_{\geq 0}(X[n]),$$

whereas the right hand side equals

$$\begin{aligned} \tau_{\leq n}((\tau_{\geq 0}(X[n]))[-n])[n] &= \tau_{\leq 0}(\tau_{\geq 0}(X[n]))[-n][n] \\ &= \tau_{\leq 0}(\tau_{\geq 0}(X[n])) \end{aligned}$$

by Lemma 2.0.6.

□

Theorem 3.0.3 (See [KW13, Theorem 4.4]). Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D . Let (X, Y, Z) be a distinguished triangle in D . The cohomological functors (Definition 3.0.1) attached to a t -structure of D induce a long exact cohomology sequence

$$\cdots \rightarrow H^{-1}(Z) \rightarrow H^0(X) \rightarrow H^0(Y) \rightarrow H^0(Z) \rightarrow H^1(X) \rightarrow \cdots$$

in $\text{Core}(D)$ (Definition 2.0.2).

Definition 3.0.4. Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D . The t -structure $(D^{\leq 0}, D^{\geq 0})$ is said to be **non-degenerate** if $\bigcap_n D^{\leq n}$ and $\bigcap_n D^{\geq n}$ each only have zero objects.

Proposition 3.0.5 ([BBD82, Proposition 1.3.7]). Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D .

(♠ **TODO: conservative system of functors**) If the t -structure $(D^{\leq 0}, D^{\geq 0})$ on D is non-degenerate (Definition 3.0.4), then the system of functors H^n is conservative, i.e. for a morphism $f : M \rightarrow N$ between objects of D , the induced maps $H^n(f) : H^n(M) \rightarrow H^n(N)$ are all isomorphisms if and only if f is an isomorphism. Moreover, an object X in D belongs to $D^{\leq 0}$ (resp. $D^{\geq 0}$) if and only if the $H^n(X)$ are zero for $n > 0$ (resp. for $n < 0$).

4. RECOLLEMENT OF t -STRUCTURE

The following from [BBD82, Section 1.4] describes a general framework for gluing t -structures on triangulated categories.

Context 4.0.1 ([BBD82, 1.4.3.]). Let D , D_U , and D_F be triangulated categories (Definition 1.0.1). Suppose that there exist exact functors (Definition 8.0.1)

$$D_F \xrightarrow{i_*} D \xrightarrow{j^*} D_U.$$

[BBD82] sets $i_! := i_*$ and $j^! := j^*$ for convenience. Assume that the following hold:

1. i_* has exact left and right adjoints, denoted by i^* and $i^!$ respectively. In other words, we have adjunctions

$$i^* \dashv i_* \dashv i^!$$

2. j^* has exact left and right adjoints, denoted by $j_!$ and j_* respectively. In other words, we have adjunctions

$$j_! \dashv j^* \dashv j_*$$

3. $j^*i_* = 0$. By adjunction, we consequently have $i^*j_! = 0$, $i^!j_* = 0$, and for all $A \in D_F$ and $B \in D_U$,

$$\mathrm{Hom}_D(j_!B, i_*A) = 0 \quad \text{and} \quad \mathrm{Hom}_D(i_*A, j_*B) = 0.$$

4. For all K in D there exists $d : i_*i^*K \rightarrow j_!j^*K[1]$ (resp. $d : j_*j^*K \rightarrow i_*i^!K[1]$), which necessarily must be unique by the above condition and Corollary 1.0.6, such that the triangle

$$\begin{aligned} j_!j^*K \rightarrow K \rightarrow i_*i^*K \xrightarrow{d} \\ (\text{resp. } i_*i^!K \rightarrow K \rightarrow j_*j^*K \xrightarrow{d}) \end{aligned}$$

is distinguished.

5. i_* , $j_!$, and j^* are fully faithful (Definition A.0.2), i.e. the adjunction morphisms $i^*i_* \rightarrow \mathrm{id} \rightarrow i^!i_*$, and $j_!j^* \rightarrow \mathrm{id} \rightarrow j_*j^*$ are isomorphisms.

This formalism is self-dual — the duality exchanges $j_!$ and j_* and i^* and $i^!$.

Definition 4.0.2 ([BBD82, 1.4.9.]). Assume Context 4.0.1. Let $(D_U^{\leq 0}, D_U^{\geq 0})$ and $(D_F^{\leq 0}, D_F^{\geq 0})$ be t -structures (Definition 2.0.2) on D_U and D_F respectively. Define

$$\begin{aligned} D^{\leq 0} &:= \{K \in D \mid j^*K \in D_U^{\leq 0} \text{ and } i^*K \in D_F^{\leq 0}\} \\ D^{\geq 0} &:= \{K \in D \mid j^*K \in D_U^{\geq 0} \text{ and } i^!K \in D_F^{\geq 0}\} \end{aligned}$$

It is a t -structure on D (Theorem 4.0.3). We might call this t -structure the one *induced from those of D_U and D_F by gluing or recollement*.

Theorem 4.0.3 ([BBD82, Théorème 1.4.10., Remarque 1.4.11.]). Assume Context 4.0.1. Let $(D_U^{\leq 0}, D_U^{\geq 0})$ and $(D_F^{\leq 0}, D_F^{\geq 0})$ be t -structures (Definition 2.0.2) on D_U and D_F respectively. The pair $(D^{\leq 0}, D^{\geq 0})$ is a t -structure (Definition 2.0.2) on D . Moreover, $(D^{\leq 0}, D^{\geq 0})$ is non-degenerate (Definition 3.0.4) if and only if $(D_U^{\leq 0}, D_U^{\geq 0})$ and $(D_F^{\leq 0}, D_F^{\geq 0})$ both are.

Proposition 4.0.4 ([BBD82, Proposition 1.4.12.]). Assume Context 4.0.1. Let $(D^{\leq 0}, D^{\geq 0})$ be a t -structure (Definition 2.0.2) on D . The following are equivalent:

1. $j_!j^* : D \rightarrow D$ is right t -exact (Definition 8.0.2).
2. $j_*j^* : D \rightarrow D$ is left t -exact (Definition 8.0.2).
3. The t -structure of D is obtained by gluing t -structures on D_U and D_F .

More explicitly, if the above hold, then the t -structures on D_U and D_F are respectively $(D_U^{\leq 0}, D_U^{\geq 0})$ and $(D_F^{\leq 0}, D_F^{\geq 0})$ uniquely given by

$$\begin{aligned} D_U^{\leq 0} &= j^*(D^{\leq 0}) \\ D_U^{\geq 0} &= j^*(D^{\geq 0}) \\ D_F^{\leq 0} &= i^*(D^{\leq 0}) \\ D_F^{\geq 0} &= i^!(D^{\geq 0}) \end{aligned}$$

5. BOUNDED DERIVED CATEGORY OF CONSTRUCTIBLE COMPLEXES OF ℓ -ADIC SHEAVES

(♠ **TODO:**) From here, let X be a scheme over a field k of characteristic $p \geq 0$ and let $\ell \neq p$ be a prime number. There is a derived category $D_c^b(X) = D_c^b(X, \overline{\mathbb{Q}}_\ell)$ of bounded complexes of constructible sheaves with $\overline{\mathbb{Q}}_\ell$ -coefficients. There is an associated six-functor formalism:

1. There are bifunctors $D_c^b(X) \rightarrow D_c^b(X) \rightarrow D_c^b(X)$ written

$$(M, N) \mapsto \mathrm{RHom}(M, N), \quad (M, N) \mapsto M \otimes N.$$

2. Given a k -morphism $f : X \rightarrow Y$, there are functors

$$f^* : D_c^b(Y) \rightarrow D_c^b(X)$$

and

$$Rf_* : D_c^b(X) \rightarrow D_c^b(Y).$$

3. If f is k -compactifiable (e.g. by virtue of Y being quasi-compact and quasi-separated and f being separated and of finite type), then there are functors $Rf_! : D_c^b(X) \rightarrow D_c^b(Y)$ and $f^! : D_c^b(Y) \rightarrow D_c^b(X)$.

These satisfy the following adjunctions (the last adjunction requires f to be k -compactifiable for $Rf_!$ to be defined):

$$\begin{aligned} \mathrm{Hom}(M \otimes N, P) &\cong \mathrm{Hom}(R, \mathrm{RHom}(N, P)) \\ \mathrm{Hom}(f^* M, N) &\cong \mathrm{Hom}(M, Rf_* N) \\ \mathrm{Hom}(Rf_! M, N) &\cong \mathrm{Hom}(M, f^! N). \end{aligned}$$

For any object $K \in D_c^b(X)$, there are the cohomology functors $K \mapsto \mathcal{H}^i(K)$. Note that each $\mathcal{H}^i(K)$ is a constructible sheaf with $\overline{\mathbb{Q}}_\ell$ -coefficients.

6. STANDARD t -STRUCTURE

Definition 6.0.1 (Standard t -structure on $D_c^b(X, \star)$). Let (R, λ) be a complete DVR such that R/λ^n is finite for all $n \geq 1$. Let E be the fraction field of R , and let K/E be an algebraic extension.

Let X be a noetherian scheme. For any coefficient $\star \in \{R, E, K\}$, the *standard t -structure on $D_c^b(X, \star)$* is given by the pair of strictly full subcategories

$$\begin{aligned} {}^{\mathrm{st}}D^{\leq 0}(X, \star) &= \{K^\bullet \in D_c^b(X, \star) : \mathcal{H}^i(K^\bullet) = 0 \text{ for all } i > 0\}, \\ {}^{\mathrm{st}}D^{\geq 0}(X, \star) &= \{K^\bullet \in D_c^b(X, \star) : \mathcal{H}^i(K^\bullet) = 0 \text{ for all } i < 0\}, \end{aligned}$$

where $\mathcal{H}^i(K^\bullet)$ denotes the i th cohomology sheaf of the complex $K^\bullet$.

For integral coefficients, this is [KW13, Definition II.6.3] (written there with $\mathfrak{o}$ for R). Theorem [KW13, II.6.4] proves that $({}^{\mathrm{st}}D^{\leq 0}(X, R), {}^{\mathrm{st}}D^{\geq 0}(X, R))$ is a t -structure on $D_c^b(X, R)$ and that the core is equivalent to the abelian category of π -adic sheaves on X via $K^\bullet \mapsto \mathcal{H}^0(K^\bullet)$.

For rational coefficients, the objects of $D_c^b(X, E)$ are those of $D_c^b(X, R)$, so the vanishing conditions on cohomology sheaves are identical. Thus $({}^{\mathrm{st}}D^{\leq 0}(X, E), {}^{\mathrm{st}}D^{\geq 0}(X, E))$ is a t -structure because it restricts the integral t -structure to the same object class.

For algebraic coefficients, every object of $D_c^b(X, K)$ descends to some finite extension L/K . The vanishing conditions on cohomology sheaves are preserved under base change along L'/L , so the subcategories $({}^{\mathrm{st}}D^{\leq 0}(X, K), {}^{\mathrm{st}}D^{\geq 0}(X, K))$ form a t -structure as the 2-direct limit of the t -structures on $D_c^b(X, L)$.

We write ${}^{\mathrm{st}}\tau_{\leq n}$ and ${}^{\mathrm{st}}\tau_{\geq n}$ for the truncation functors of this t -structure. The lower truncation functor ${}^{\mathrm{st}}\tau_{\leq n}$ is the Deligne truncation functor (see Lemma [KW13, II.6.1]).

Claim 6.0.2. There is a natural isomorphism

$$H^n(K) = {}^{\text{st}}\tau_{\leq n} {}^{\text{st}}\tau_{\geq n} K \cong \mathcal{H}^n(K)$$

for $K \in D_c^b(X, \overline{\mathbb{Q}}_\ell)$ where H^n denotes the n th cohomology functor (Definition 3.0.1) with respect to the standard t -structure. This follows from [KW13, Remark p.104] and the equivalence of the core with the abelian category of π -adic sheaves.

7. PERVERSE t -STRUCTURE

Definition 7.0.1 (Dualizing complex for Verdier duality on a noetherian scheme with torsion coefficients, cf. [?, Theorem 8.4.4]). Let S be a noetherian scheme. Let A be a noetherian torsion ring. Consider sheaves of A -modules on the (small) étale site of S .

The *dualizing complex for Verdier duality on S with coefficients in A* , denoted by $\mathbb{D}_S^\bullet$, is the object in $D^+(S_{\text{ét}}, A)$ characterized by the following property: for every S -scheme X of finite type with structure morphism $f : X \rightarrow S$, the dualizing complex on X relative to S satisfies

$$\mathbb{D}_{X/S}^\bullet = Rf^! \mathbb{D}_S^\bullet$$

where $Rf^! : D^+(S_{\text{ét}}, A) \rightarrow D^+(X_{\text{ét}}, A)$ is the exceptional inverse image functor for torsion coefficients along f .

Concretely, $\mathbb{D}_S^\bullet$ is the constant sheaf A_S concentrated in degree 0.

Definition 7.0.2 (Dualizing complex for Verdier duality on a noetherian scheme with integral coefficients, cf. [?, Theorem 8.4.4]). Let (R, λ) be a complete DVR such that R/λ^n is finite for all $n \geq 1$. Let S be a noetherian scheme. Consider sheaves on the (small) étale site of S .

The *dualizing complex for Verdier duality on S with integral coefficients*, denoted by $\mathbb{D}_S^\bullet(R)$, is the object in $D_c^b(S, R)$ characterized by the following property: for every S -scheme X of finite type with structure morphism $f : X \rightarrow S$, the dualizing complex on X relative to S with integral coefficients satisfies

$$\mathbb{D}_{X/S}^\bullet(R) = Rf^! \mathbb{D}_S^\bullet(R)$$

where $Rf^! : D_c^b(S, R) \rightarrow D_c^b(X, R)$ is the exceptional inverse image functor on bounded constructible complexes with integral coefficients along f .

Concretely, $\mathbb{D}_S^\bullet(R)$ is the constant inverse system $(R/\lambda^n)_n$ concentrated in degree 0.

Definition 7.0.3 (Dualizing complex for Verdier duality on a noetherian scheme with rational coefficients, cf. [?, Theorem 8.4.4]). Let (R, λ) be a complete DVR such that R/λ^n is finite for all $n \geq 1$. Let E be the fraction field of R . Let S be a noetherian scheme. Consider sheaves on the (small) étale site of S .

The *dualizing complex for Verdier duality on S with rational coefficients*, denoted by $\mathbb{D}_S^\bullet(E)$, is the object in $D_c^b(S, E)$ characterized by the following property: for every S -scheme X of

finite type with structure morphism $f : X \rightarrow S$, the dualizing complex on X relative to S with rational coefficients satisfies

$$\mathbb{D}_{X/S}^\bullet(E) = Rf^! \mathbb{D}_S^\bullet(E)$$

where $Rf^! : D_c^b(S, E) \rightarrow D_c^b(X, E)$ is the exceptional inverse image functor on bounded constructible complexes with rational coefficients along f .

Concretely, $\mathbb{D}_S^\bullet(E)$ is obtained from the dualizing complex for Verdier duality on S with integral coefficients (Definition 7.0.2) $\mathbb{D}_S^\bullet(R)$ by localization at λ . Thus $\mathbb{D}_S^\bullet(E)$ is the constant inverse system $(E)_n$ concentrated in degree 0, where each $E \cong R[\lambda^{-1}]$.

Definition 7.0.4 (Dualizing complex for Verdier duality on a noetherian scheme with algebraic coefficients, cf. [?, Theorem 8.4.4]). Let (R, λ) be a complete DVR such that R/λ^n is finite for all $n \geq 1$. Let E be the fraction field of R , and let K/E be an algebraic extension. Let S be a noetherian scheme. Consider sheaves on the (small) étale site of S .

The category $D_c^b(S, K)$ is the 2-direct limit of $D_c^b(S, L')$ as L' ranges over finite extensions of E inside K . The *dualizing complex for Verdier duality on S with algebraic coefficients*, denoted by $\mathbb{D}_S^\bullet(K)$, is the object in $D_c^b(S, K)$ characterized by the following property: for every S -scheme X of finite type with structure morphism $f : X \rightarrow S$, the dualizing complex on X relative to S with algebraic coefficients satisfies

$$\mathbb{D}_{X/S}^\bullet(K) = Rf^! \mathbb{D}_S^\bullet(K)$$

where $Rf^! : D_c^b(S, K) \rightarrow D_c^b(X, K)$ is the colimit of the exceptional inverse image functors $Rf^! : D_c^b(S, L') \rightarrow D_c^b(X, L')$ for finite extensions L'/E inside K .

Concretely, $\mathbb{D}_S^\bullet(K)$ is the image of $\mathbb{D}_S^\bullet(E)$ in $D_c^b(S, K)$ under the transition functor $D_c^b(S, E) \rightarrow D_c^b(S, K)$.

Definition 7.0.5 (Verdier dual of a constructible complex with algebraic coefficients, cf. [?, Theorem 8.4.4]). Let (R, λ) be a complete DVR such that R/λ^n is finite for all $n \geq 1$. Let E be the fraction field of R , and let K/E be an algebraic extension. Let S be a noetherian scheme. Consider sheaves on the (small) étale sites of schemes over S .

Let X be an S -scheme of finite type. For any complex $K' \in D_c^b(X, K)$, the *Verdier dual of K' on X relative to S with algebraic coefficients* is

$$\mathbb{D}_X(K') := R\mathcal{H}om_K(K', \mathbb{D}_{X/S}^\bullet(K))$$

where $\mathbb{D}_{X/S}^\bullet(K)$ is the dualizing complex on X relative to S with algebraic coefficients (Definition 7.0.4), and $R\mathcal{H}om_K : D_c^b(X, K)^{\text{op}} \times D_c^b(X, K) \rightarrow D_c^b(X, K)$ is the derived internal Hom functor obtained as the colimit of $R\mathcal{H}om_{L'}$ over finite extensions L'/E inside K (derived internal Hom on bounded constructible complexes with algebraic coefficients).

Definition 7.0.6 (Self-dual (middle) perverse t -structure with algebraic coefficients, cf. [KW13, Section III.3]). Let (R, λ) be a complete DVR such that R/λ^n is finite for all $n \geq 1$. Let E be the fraction field of R , and let K/E be an algebraic extension. Let X be a scheme of finite type over a field k that is separably closed or finite. Consider sheaves on the (small) étale site of X .

The category $D_c^b(X, K)$ is a triangulated category (Definition 1.0.1) (??). Let $\mathbb{D}_X(-) : D_c^b(X, K) \rightarrow D_c^b(X, K)$ denote the Verdier dual functor (Definition 7.0.5) on X relative to $\text{Spec } k$.

The *self-dual (middle) perverse t -structure on $D_c^b(X, K)$* is the pair of strictly full subcategories $({}^pD^{\leq 0}(X, K), {}^pD^{\geq 0}(X, K))$ of $D_c^b(X, K)$ defined by

$$\begin{aligned} {}^pD^{\leq 0}(X, K) &:= \{K' \in D_c^b(X, K) : \dim \text{Supp}(\mathcal{H}^{-i}(K')) \leq i \text{ for all } i \in \mathbb{Z}\}, \\ {}^pD^{\geq 0}(X, K) &:= \{K' \in D_c^b(X, K) : \dim \text{Supp}(\mathcal{H}^{-i}(\mathbb{D}_X(K'))) \leq i \text{ for all } i \in \mathbb{Z}\}, \end{aligned}$$

(Definition 7.0.5)

This is a t -structure (Definition 2.0.2) on $D_c^b(X, K)$ (Theorem 7.0.7). Its core (Definition 2.0.2), denoted by $\text{Perv}(X, K)$, is called the *category of perverse sheaves with algebraic coefficients on X* .

We say that $B \in D_c^b(X, K)$ is *semiperverse* if $B \in {}^pD^{\leq 0}(X)$.

Theorem 7.0.7 (See [KW13, Section III.3]). Let (R, λ) be a complete DVR such that R/λ^n is finite for all $n \geq 1$. Let E be the fraction field of R , and let K/E be an algebraic extension. Let X be a scheme of finite type over a field k that is separably closed or finite. Consider sheaves on the (small) étale site of X .

The perverse t -structure (Definition 7.0.6) on $D_c^b(X, K)$ is indeed a t -structure (Definition 2.0.2).

Write $\text{Perv}(X)$ for the core of the perverse t -structure and write ${}^p\mathcal{H}^i : D_c^b(X) \rightarrow \text{Perv}(X)$ for the i th cohomology object with respect to this t -structure. Moreover, since the perverse t -structure is a genuine t -structure, we may write ${}^pD^{\leq n}, {}^pD^{\geq n}, {}^p\tau_{\geq n}, {}^p\tau_{\leq n}$, etc. for the relevant subcategories of $D_c^b(X)$ and the truncation functors.

Lemma 7.0.8. For $B \in D_c^b(X)$,

$$\begin{aligned} B \in {}^pD^{\leq n}(X) &\Leftrightarrow \dim \text{Supp}(\mathcal{H}^{-i}B) \leq i + n \quad \text{for all } i \in \mathbb{Z} \\ B \in {}^pD^{\geq n}(X) &\Leftrightarrow \dim \text{Supp}(\mathcal{H}^{-i}DB) \leq i - n \quad \text{for all } i \in \mathbb{Z} \end{aligned}$$

Proof. By definition, ${}^pD^{\leq n}(X) = {}^pD^{\leq 0}(X)[-n]$, so $B \in {}^pD^{\leq n}(X)$ if and only if $B[n] \in {}^pD^{\leq 0}(X)$. In turn, this is equivalent to

$$\dim \text{Supp}(\mathcal{H}^{-i}(B[n])) \leq i \quad \text{for all } i \in \mathbb{Z}$$

or to

$$\dim \text{Supp}(\mathcal{H}^{-i+n}B) \leq i \quad \text{for all } i \in \mathbb{Z}$$

by Lemma 3.0.2. Changing indices ($j = i - n$), this is equivalent to

$$\dim \text{Supp}(\mathcal{H}^{-j}B) \leq j + n \quad \text{for all } j \in \mathbb{Z}$$

as desired.

Similarly, $B \in {}^pD^{\geq n}(X) = {}^pD^{\geq 0}(X)[-n]$ if and only if $B[n] \in {}^pD^{\geq 0}(X)$, which is equivalent to $D(B[n]) \in {}^pD^{\leq 0}(X)$. Because $D(B[n]) \cong (DB)[-n]$, this means $(DB)[-n] \in {}^pD^{\leq 0}(X)$, which translates to

$$\dim \operatorname{Supp}(\mathcal{H}^{-i}((DB)[-n])) \leq i \quad \text{for all } i \in \mathbb{Z}$$

Using the shift property $\mathcal{H}^{-i}((DB)[-n]) \cong \mathcal{H}^{-i-n}(DB)$, we obtain

$$\dim \operatorname{Supp}(\mathcal{H}^{-i-n}DB) \leq i \quad \text{for all } i \in \mathbb{Z}$$

Setting $j = i + n$ yields $\dim \operatorname{Supp}(\mathcal{H}^{-j}DB) \leq j - n$, completing the proof. $\square$

Lemma 7.0.9. (♠ TODO:) For nonzero $B \in D_c^b(X)$,

1. $B \notin {}^pD^{\geq n}(X)$ for all $n \ll 0$.
2. $B \notin {}^pD^{\leq n}(X)$ for all $n \gg 0$.

Proof. The condition $B \in {}^pD^{\geq n}(X)$ is equivalent to the statement

$$\dim \operatorname{Supp}(\mathcal{H}^{-i}B) \leq i + n \quad \text{for all } i \in \mathbb{Z}$$

by Lemma 7.0.8. Since B is assumed to be nonzero, at least one of the $\mathcal{H}^{-i}B$ must be nonzero; the inequality $\dim \operatorname{Supp}(\mathcal{H}^{-i}B) \leq i + n$ cannot hold whenever $n < -i$. Therefore, B cannot be an object of ${}^pD^{\geq n}(X)$ for any such n . Similarly, $B \notin {}^pD^{\leq n}(X)$ for all $n \gg 0$ as well. $\square$

Corollary 7.0.10. (♠ TODO:) The perverse t -structure is non-degenerate. In particular, for $B \in D_c^b(X)$, ${}^p\mathcal{H}^i(B) = 0$ for all $i \in \mathbb{Z}$ if and only if $B = 0$ and

$$\begin{aligned} {}^pD^{\leq n}(X) &= \{B \in D_c^b(X) : {}^p\mathcal{H}^i(B) = 0 \text{ for all } i > n\} \\ {}^pD^{\geq n}(X) &= \{B \in D_c^b(X) : {}^p\mathcal{H}^i(B) = 0 \text{ for all } i < n\}. \end{aligned}$$

Proof. This is immediate from Lemma 7.0.9, the definition of a non-degenerate t -structure, and Proposition 3.0.5. $\square$

Theorem 7.0.11 (See [KW13, Corollaries III.5.4, III.5.7]). (♠ TODO:) $\operatorname{Perv}(X)$ is artinian and noetherian. Moreover, $B \in \operatorname{Perv}(X)$ is simple if and only if it is of the form $B = j_{!*}\mathcal{G}[d]$ for a locally closed embedding $j : U \rightarrow X$ from an irreducible smooth d -dimensional subscheme U of dimension d and a lisse $\overline{\mathbb{Q}}_\ell$ -sheaf $\mathcal{G}$.

Proposition 7.0.12 (See [KW13, Lemma III.5.13]). If $\dim(X) \leq d$, then ${}^pD^{\geq 0}(X, \overline{\mathbb{Q}}_\ell)$ is contained in ${}^{\operatorname{st}}D^{\geq -d}(X, \overline{\mathbb{Q}}_\ell)$.

8. EXACTNESS PROPERTIES FOR THE PERVERSE t -STRUCTURE UNDER DIRECT IMAGES AND INVERSE IMAGES

We discuss some conditions under which the derived functors $Rf_*, Rf_!, f^*, f^!$ (or variants thereof) preserve perversity.

Definition 8.0.1. A functor $T : D_1 \rightarrow D_2$ between triangulated categories (Definition 1.0.1) is called *exact* if it is additive, is translation preserving, and transforms distinguished triangles (Definition 1.0.1) to distinguished triangles. Oftentimes, a *morphism between triangulated categories* or a *triangulated functor* refers to an exact functor between triangulated categories.

Definition 8.0.2. Let $T : D_1 \rightarrow D_2$ be an exact functor (Definition 8.0.1) between triangulated categories (Definition 1.0.1) with t -structure (Definition 2.0.2). Such a functor is called *t -left exact* if $T(D_1^{\leq 0}) \subset D_2^{\leq 0}$, *t -right exact* if $T(D_1^{\geq 0}) \subset D_2^{\geq 0}$, and *t -exact* if T is both t -left and t -right exact.

Lemma 8.0.3 (cf. [Kat96, Section 2.1], [Kat90, Section 8.1],). Let $f : X \rightarrow Y$ be a morphism of finite type schemes over a field k of characteristic distinct from ℓ .

1. ([BBD82, Théorème 4.1.1], cf. [DA73, Exposé XIV, Théorème], [KW13, Theorem III.6.1]) If $f : X \rightarrow Y$ is affine, then $Rf_* : D_c^b(X, \overline{\mathbb{Q}}_\ell) \rightarrow D_c^b(Y, \overline{\mathbb{Q}}_\ell)$ is right t -exact with respect to the perverse t -structures on $D_c^b(X, \overline{\mathbb{Q}}_\ell)$ and $D_c^b(Y, \overline{\mathbb{Q}}_\ell)$, i.e. restricts to a functor ${}^pD^{\leq 0}(X) \rightarrow {}^pD^{\leq 0}(Y)$, and $Rf_! : D_c^b(X, \overline{\mathbb{Q}}_\ell) \rightarrow D_c^b(Y, \overline{\mathbb{Q}}_\ell)$ is left t -exact. In particular, Rf_* preserves semiperversity.
2. (See [BBD82, Proposition 2.2.5]) If $f : X \rightarrow Y$ is a quasi-finite, then $Rf_!$ and f^* are right t -exact, and $f^!$ and Rf_* are left t -exact. In particular, $Rf_!$ and f^* preserve semiperversity.

Proposition 8.0.4. Let $f : X \rightarrow Y$ be a morphism of finite type schemes over a field k of characteristic distinct from ℓ .

1. If f is quasi-finite and affine, then $Rf_*M, Rf_!M \in \text{Perv}(Y)$ for all $M \in \text{Perv}(X)$ [KW13, Corollary III.6.2].
2. If f is equidimensional of relative dimension d and smooth, then

$$f^*[d](M) = f^!-d(M) \in \text{Perv}(X)$$

for all $M \in \text{Perv}(Y)$ [KW13, Theorem III.7.2].

APPENDIX A. MISCELLANEOUS DEFINITIONS

Definition A.0.1. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ be functors.

An *adjunction between F and G* consists of two natural transformations: $\eta : \text{Id}_{\mathcal{C}} \Rightarrow GF$ (the *unit*), and $\varepsilon : FG \Rightarrow \text{Id}_{\mathcal{D}}$ (the *counit*)

These must satisfy the triangle identities: For every object $X \in \mathcal{C}$ and $Y \in \mathcal{D}$,

$$\varepsilon_{FX} \circ F(\eta_X) = \text{id}_{FX}$$

$$G(\varepsilon_Y) \circ \eta_{GY} = \text{id}_{GY}.$$

In diagrammatic form, the triangle identities assert that the following are commutative diagrams:

$$\begin{array}{ccc}
F(X) & \xrightarrow{F(\eta_X)} & FGF(X) \\
\searrow \text{id}_{F(X)} & & \downarrow \varepsilon_{F(X)} \\
& & F(X)
\end{array}
\quad
\begin{array}{ccc}
G(Y) & \xrightarrow{\eta_{G(Y)}} & GFG(Y) \\
\searrow \text{id}_{G(Y)} & & \downarrow G(\varepsilon_Y) \\
& & G(Y)
\end{array}$$

We say that F is a **left adjoint to G** and G is a **right adjoint to F** (written $F \dashv G$).

In the case that $\mathcal{C}$ and $\mathcal{D}$ are locally small categories (or U -locally small categories if a universe U is available), we have an adjunction $F \dashv G$ if and only if for every object X in $\mathcal{C}$ and Y in $\mathcal{D}$ there is a natural isomorphism

$$\text{Hom}_{\mathcal{D}}(F(X), Y) \cong \text{Hom}_{\mathcal{C}}(X, G(Y))$$

that is natural in both X and Y . More precisely, there are two functors

$$\mathcal{C}^{\text{op}} \times \mathcal{D} \rightarrow \mathbf{Sets}$$

given by

$$\begin{aligned}
(X, Y) &\mapsto \text{Hom}_{\mathcal{D}}(F(X), Y) \\
(X, Y) &\mapsto \text{Hom}_{\mathcal{C}}(X, G(Y))
\end{aligned}$$

respectively, and there is a natural isomorphism between them.

In this case, the **unit of the adjunction** is the natural transformation $\eta : \text{Id}_{\mathcal{C}} \Rightarrow GF$ such that,

1. for every $X \in \mathcal{C}$, the morphism $\eta_X : X \rightarrow GF(X)$ (each called a **unit morphism**) in $\mathcal{C}$ is obtained as the image of $\text{id}_{F(X)}$ via the adjoint isomorphism

$$\text{Hom}_{\mathcal{D}}(F(X), F(X)) \cong \text{Hom}_{\mathcal{C}}(X, GF(X)).$$

2. for every $Y \in \mathcal{D}$, the morphism $\epsilon_Y : FG(Y) \rightarrow Y$ (each called a **counit morphism**) in $\mathcal{D}$ is obtained as the image of $\text{id}_{G(Y)}$ via the adjoint isomorphism

$$\text{Hom}_{\mathcal{C}}(G(Y), G(Y)) \cong \text{Hom}_{\mathcal{D}}(FG(Y), Y).$$

Definition A.0.2. Let $\mathcal{C}$ and $\mathcal{D}$ be (large) categories. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor.

1. F is called **full** if for every pair of objects $x, y \in \text{Ob}(\mathcal{C})$, the induced rule/assignment/class function

$$F_{x,y} : \text{Hom}_{\mathcal{C}}(x, y) \rightarrow \text{Hom}_{\mathcal{D}}(F(x), F(y))$$

on Hom-collections is “surjective”, i.e. for all morphisms $g : F(x) \rightarrow F(y)$, there exists some morphism $f : x \rightarrow y$ such that $F(f) = g$.

2. F is called **faithful** if for every pair of objects $x, y \in \text{Ob}(\mathcal{C})$, the induced class function (assignment)

$$F_{x,y} : \text{Hom}_{\mathcal{C}}(x, y) \rightarrow \text{Hom}_{\mathcal{D}}(F(x), F(y))$$

on Hom-collections is “injective”, i.e., for any morphisms $f_1, f_2 \in \text{Hom}_{\mathcal{C}}(x, y)$, if $F(f_1) = F(f_2)$ in $\text{Hom}_{\mathcal{D}}(F(x), F(y))$, then $f_1 = f_2$.

3. F is called **fully faithful** if it is both full and faithful.

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